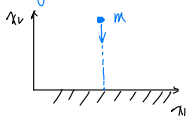
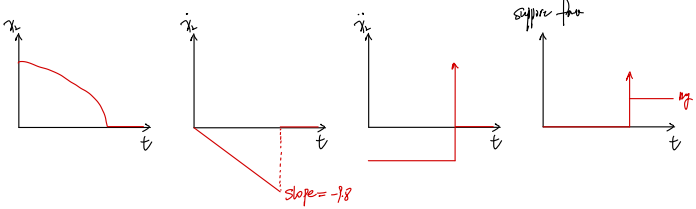
- Contact

Falling Ball



$\phi(q) = [0,1]\, q$ $\quad \phi(q) = 0$ means contact



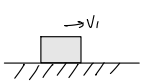
$L(q,\dot{q}) = \frac{1}{2} m \|\dot{q}\|^2 - mg[0,1]\, q$

$$\min_{q[1..N]} \sum_{k=1}^{N-1} h L\left(\frac{q_{k+1}+q_k}{2}, \frac{q_{k+1}-q_k}{h}\right) = \sum_{k=1}^{N-1} L_d(q_k, q_{k+1})$$

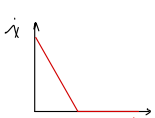
s.t. $\phi(q_k) \geq 0 \quad : n_k \qquad \forall k = 2 \cdots N$

$$\begin{cases} \frac{\partial}{\partial q_k} L_d(q_{k-1}, q_k) + \frac{\partial}{\partial q_k} L_d(q_k, q_{k+1}) = \frac{\partial \phi}{\partial q_k}^T n_k & \text{DEL with constraints} \\ \phi(q_k) \geq 0 & \text{Stay above ground} \\ n_k \geq 0 & \text{Support force} \\ \phi(q_k) \cdot n_k = 0 & \text{only have support force at contact} \end{cases}$$

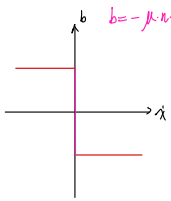
- Coulomb Friction



a sliding object come to rest due to friction







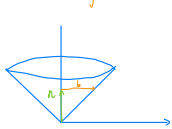
$b = -\mu \cdot n \cdot \text{sign}(\dot{x})$ where n is the support force, μ is the friction coefficient

when the obj is not sliding
the friction can be anything balancing the external force

maximum dissipation principle : the true friction maximally reduce the kinetic energy

$$\min_b \; \dot{x}^T b$$
$$\text{s.t. } \|b\|_2 \leq \mu n$$

friction cone



$\dot{T} = \frac{d}{dt}\left(\frac{1}{2}\dot{q}^T M(q)\dot{q}\right)$

$= (M(q)\dot{q})^T\ddot{q} + \frac{1}{2}\dot{q}^T\left(\frac{\partial M}{\partial q}\dot{q}\right)\dot{q}$ be is a force, only effects $\ddot{q}$

$\Downarrow$

$\min_b \quad \dot{q}^T M(q)\ddot{q}$

s.t. $\|b\|_2 \le \mu n$

$\Downarrow$

From $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = Dr(q)^T b$ (if only one particle)

$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = Dr(q)^T b$

$\Downarrow$

$\min_b \quad \dot{q}^T Dr(q)^T b = \langle Dr(q)\dot{q}, b\rangle$ the friction fight against velocity

s.t. $\|b\|_2 \le \mu n$

$\Downarrow$

$\min_b \quad \frac{(q_{k+1}-q_k)^T}{h} Dr(q_k)^T b$

s.t. $\|b\|_2 \le \mu n \quad : \lambda$

$Dr(q_k)\frac{(q_{k+1}-q_k)}{h} + \lambda\frac{b}{\|b\|_2} = 0 \qquad \lambda = \|Dr(q_k)\frac{(q_{k+1}-q_k)}{h}\|_2 = \|v\|_2$

$\lambda \ge 0$

$\|b\|_2 \le \mu n$

$\lambda(\|b\|_2 - \mu n) = 0$

if $\lambda \ne 0$ $\|b\|_2 - \mu n = 0$, taking the maximum friction

if $\|b\|_2 < \mu n$, not taking friction $\lambda = \|v\|_2 = 0$, obj not moving

2. Relaxation of Coulomb Friction

$\min_{\lambda_t} \quad v^T\lambda_t$ s.t. $\|\lambda_t\|_2 \le \mu\lambda_n$ ⟹ 2 Approximation ⟹ $\min_{\lambda_t} \quad v^T\lambda_t$ s.t. $\|\lambda_t\|_1 \le \mu\lambda_n$

theorem

$\min_{\lambda_t} \quad v^T\lambda_t$ s.t. $\|\lambda_t\|_1 \le \mu\lambda_n$ $\iff$ $0 \le \begin{bmatrix}\lambda_t^+ \\ \lambda_t^- \\ \gamma\end{bmatrix} \perp \begin{bmatrix} V+\gamma 1 \\ -V+\gamma 1 \\ \mu\lambda_n - 1^T(\lambda_t^+ + \lambda_t^-)\end{bmatrix} \ge 0$

$L = v^T\lambda_t + \gamma[\|\lambda_t\|_1 - \mu\lambda_n]$

① if $\lambda_t[i] > 0$ $\partial L/\partial\lambda_t[i] = V_i + \gamma = 0$

② if $\lambda_t[i] < 0$ $\partial L/\partial\lambda_t[i] = V_i - \gamma = 0$

③ if $\lambda_t[i] = 0$ $0 \in \partial L/\partial\lambda_t[i] = V_i + \gamma[-1,1]$

④ $\gamma[\|\lambda_t\|_1 - \mu\lambda_n] = 0$

① if $\gamma = 0$, in both case we have

$v \geq 0$, λ_c can be anything s.t. $\|\lambda_c\|_1 = \|\lambda_c^+ - \lambda_c^-\|_1 \leq \mathbf{1}^T\lambda_c^+ + \mathbf{1}^T\lambda_c^- \leq \mu\lambda_n$

② if $\gamma > 0$

($\Rightarrow$) suppose (λ_c, γ) satisfies the KKT condition

let $\lambda_c^+ = \max(\lambda_c, 0)$ $\lambda_c^- = \max(-\lambda_c, 0)$

then $\|\lambda_c\|_1 = \mathbf{1}^T\lambda_c^+ + \mathbf{1}^T\lambda_c^-$, $(\lambda_c^+, \lambda_c^-, \gamma)$ satisfies the LCP

($\Leftarrow$) suppose $(\lambda_c^+, \lambda_c^-, \gamma)$ satisfies the LCP

if $\lambda_c^+[i] > 0$, $V_i + \gamma = 0$ $V_i = -\gamma < 0$, $-V_i + \gamma > 0$, $\lambda_c^-[i] = 0$

if $\lambda_c^-[i] > 0$, $-V_i + \gamma = 0$ $V_i = \gamma > 0$, $V_i + \gamma > 0$, $\lambda_c^+[i] = 0$

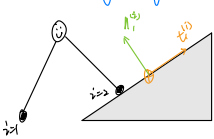
thus λ_c^+ and λ_c^- are complementary

then $\|\lambda_c^+ - \lambda_c^-\|_1 = \mathbf{1}^T\lambda_c^+ + \mathbf{1}^T\lambda_c^-$

then $(\lambda_c^+ - \lambda_c^-, \gamma)$ satisfies the KKT condition

· Time-Stepping LCP

(assuming single contact point for each particle)



① $t_{i2}^{(j)}$ pointing out the screen

the normal space at i is $N^{(j)} = \left[n_i^{(j)}\right] = \frac{d\phi^{(j)T}}{dr^{(j)}}$

the tangent space at i is $T^{(j)} = \left[t_{i1}^{(j)}\ t_{i2}^{(j)}\right]$

} assume $\frac{d\phi^{(j)}}{dr^{(j)}}$ is unit norm, $\left[n_i^{(j)}\ t_{i1}^{(j)}\ t_{i2}^{(j)}\right]$ is orthonormal

the tangent velocity at i is $V_t^{(j)} = T^{(j)T} Dr^{(j)}(q)\dot{q}$ in end-effector coordinate

the friction force at i is $T^{(j)}\lambda_t^{(j)}$ in $T^{(j)}$ coordinate — in world coordinate

the contact force at i is $N^{(j)}\lambda_n^{(j)}$ in world coordinate

Lagrange Dynamics

$$M(q)\ddot{q} + C(q,\dot{q}) = \sum_{j=1}^{N} Dr^{(j)}(q)^T \begin{bmatrix} N^{(j)} & T^{(j)} \end{bmatrix} \begin{bmatrix} \lambda_n^{(j)} \\ \lambda_t^{(j)} \end{bmatrix}$$

$$= \sum_{j=1}^{N} Dr^{(j)}(q)^T \begin{bmatrix} N^{(j)} & T^{(j)} & -T^{(j)} \end{bmatrix} \begin{bmatrix} \lambda_n^{(j)} \\ \lambda_t^{(j)+} \\ \lambda_t^{(j)-} \end{bmatrix}$$

$$= \sum_{j=1}^{N} \underbrace{J_n^{(j)T}\lambda_n^{(j)}}_{} + \underbrace{J_t^{(j)T}\left(\lambda_t^{(j)+} - \lambda_t^{(j)-}\right)}_{}$$

normal jacobian $J_n^{(j)} = N^{(j)T} Dr^{(j)}(q) \in 1 \times \dim(q)$ $T^{(j)T} Dr^{(j)}(q)$ tangent jacobian $\in 2 \times \dim(q)$

when q change a little, how much $r^{(j)}$ change in normal direction

Contact Complementarity $0 \le \phi^{(i)}(r^{(i)}) \perp \lambda_n^{(i)} \ge 0$

Friction Complementarity $0 \in \begin{bmatrix} \lambda_n^{(i)} \\ \lambda_{t_1}^{(i)+} \\ \lambda_{t_2}^{(i)-} \end{bmatrix} \perp \begin{bmatrix} V_{t_1}^{(i)} + \gamma^{(i)} 1 \\ -V_{t_1}^{(i)} + \gamma^{(i)} 1 \\ \mu^{(i)} \lambda_n^{(i)} - 1^T (\lambda_{t_1}^{(i)+} + \lambda_{t_2}^{(i)-}) \end{bmatrix} \ge 0$

① Discrete Lagrange dynamics (forward in time)

$$M(q_k)\left(\frac{\dot{q}_{k+1} - \dot{q}_k}{h}\right) + C(q_k, \dot{q}_k) = \sum_{i=1}^{N} \left[J_n^{(i)T} \lambda_n^{(i)} + J_{t_1}^{(i)T} \lambda_{t_1}^{(i)+} - J_{t_2}^{(i)T} \lambda_{t_2}^{(i)-} \right]$$

$$\dot{q}_{k+1} = \dot{q}_k + h M(q_k)^{-1} \left[\sum_{i=1}^{N} \left(J_n^{(i)T} \lambda_n^{(i)} + J_{t_1}^{(i)T} \lambda_{t_1}^{(i)+} - J_{t_2}^{(i)T} \lambda_{t_2}^{(i)-} \right) - C(q_k, \dot{q}_k) \right]$$

② Discretize q (backward in time)

$$q_{k+1} = q_k + h \dot{q}_{k+1}$$

③ Linearize ϕ (backward in time)

$$\frac{\phi^{(i)}(r^{(i)}(q_{k+1}))}{\| \nabla \phi^{(i)}(r^{(i)}(q_{k+1})) \|_2} \approx \frac{\phi^{(i)}(r^{(i)}(q_k))}{\| \nabla \phi^{(i)}(r^{(i)}(q_k)) \|_2} + h \underbrace{\underbrace{\frac{d\phi^{(i)}/dr^{(i)}(q_k)}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|_2}}_{N^{(i)T}} Dr^{(i)}(q_k)}_{J_n^{(i)}} \dot{q}_{k+1}$$

$$\approx \frac{\phi^{(i)}(r^{(i)}(q_k))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|_2} + h J_n^{(i)} \left\{ \dot{q}_k + h M(q_k)^{-1} \left[\sum_{i=1}^{N} \left(J_n^{(i)T} \lambda_n^{(i)} + J_{t_1}^{(i)T} \lambda_{t_1}^{(i)+} - J_{t_2}^{(i)T} \lambda_{t_2}^{(i)-} \right) - C(q_k, \dot{q}_k) \right] \right\}$$

$$\approx \frac{\phi^{(i)}(r^{(i)}(q_k))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|_2} + \underbrace{h J_n^{(i)} \dot{q}_k}_{\text{1st order, due to velocity}} + \underbrace{h^2 J_n^{(i)} M(q_k)^{-1}}_{\text{2nd-order, due to external force}} \left[\sum_{i=1}^{N} \left(J_n^{(i)T} \lambda_n^{(i)} + J_{t_1}^{(i)T} \lambda_{t_1}^{(i)+} - J_{t_2}^{(i)T} \lambda_{t_2}^{(i)-} \right) - C(q_k, \dot{q}_k) \right]$$

$$\frac{1}{h} \frac{\phi^{(i)}(r^{(i)}(q_{k+1}))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|_2} = \frac{1}{h} \frac{\phi^{(i)}(r^{(i)}(q_k))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|_2} + J_n^{(i)} \dot{q}_k - h J_n^{(i)} M(q_k)^{-1} C(q_k, \dot{q}_k) + h \sum_{i=1}^{N} \left[\underbrace{J_{nn}^{(ij)}}_{} \lambda_n^{(j)} + J_{nt}^{(ij)} \lambda_{t_1}^{(j)+} - \underbrace{J_{nt}^{(ij)}}_{} \lambda_{t_2}^{(j)-} \right]$$

$$J_{nn}^{(ij)} = J_n^{(i)} M(q_k)^{-1} J_n^{(j)T} \qquad J_{nt}^{(ij)} = J_n^{(i)} M(q_k)^{-1} J_t^{(j)T}$$

④ tangent velocity (backward in time)

$$V_t^{(i)} = T^{(i)T} Dr^{(i)}(q_k) \dot{q}_{k+1}$$

$$= J_t^{(i)} \dot{q}_{k+1}$$

$$= J_t^{(i)} \left\{ \dot{q}_k + h M(q_k)^{-1} \left[\sum_{i=1}^{N} \left(J_n^{(i)T} \lambda_n^{(i)} + J_{t_1}^{(i)T} \lambda_{t_1}^{(i)+} - J_{t_2}^{(i)T} \lambda_{t_2}^{(i)-} \right) - C(q_k, \dot{q}_k) \right] \right\}$$

$$= J_t^{(i)} \dot{q}_k - h J_t^{(i)} M(q_k)^{-1} C(q_k, \dot{q}_k) + h \left[\sum_{i=1}^{N} J_{tn}^{(ij)} \lambda_n^{(j)} + J_{tt}^{(ij)} \lambda_{t_1}^{(j)+} - J_{tt}^{(ij)} \lambda_{t_2}^{(j)-} \right]$$

$$0 \le \begin{bmatrix} \lambda_n^{(i)} \\ \lambda_t^{(i)+} \\ \lambda_t^{(i)-} \\ \gamma^{(i)} \end{bmatrix} \perp \begin{bmatrix} \frac{\phi^{(i)}(r^{(i)}(q_{k+1}))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|} \\ V_t^{(i)} + \gamma^{(i)} 1 \\ -V_t^{(i)} + \gamma^{(i)} 1 \\ \mu^{(i)} \lambda_n^{(i)} - I^T(\lambda_t^{(i)+} + \lambda_t^{(i)-}) \end{bmatrix} \ge 0$$

$$0 \le \begin{bmatrix} \lambda_n^{(i)} \\ \lambda_n^{(\omega)} \end{bmatrix} \perp \begin{bmatrix} \frac{1}{h} \frac{\phi^{(i)}(r^{(i)}(q_{k+1}))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|} \\ \cdots (*) \end{bmatrix} = \begin{matrix} \downarrow \\ N\times 1 \\ \uparrow \end{matrix} h \begin{bmatrix} J_{nn}^{(ii)} & J_{nn}^{(i\omega)} \\ J_{nn}^{(\omega i)} & J_{nn}^{(\omega\omega)} \end{bmatrix} \quad \begin{matrix} \rightarrow N\times 1 \leftarrow \\ h \begin{bmatrix} J_{ne}^{(ii)} & J_{ne}^{(i\omega)} \\ J_{ne}^{(\omega i)} & J_{ne}^{(\omega\omega)} \end{bmatrix} \end{matrix} \quad \begin{matrix} \rightarrow N\times 2 \leftarrow \\ -h \begin{bmatrix} J_{ne}^{(ii)} & J_{ne}^{(i\omega)} \\ J_{ne}^{(\omega i)} & J_{ne}^{(\omega\omega)} \end{bmatrix} \end{matrix} \quad \begin{matrix} \leftarrow N\times 4 \\ \end{matrix} \quad \begin{bmatrix} 0 \\ \end{bmatrix}$$

$$\begin{matrix} \downarrow \\ N\times 2 \\ \uparrow \end{matrix} \begin{bmatrix} V_t^{(i)} + \gamma^{(i)} 1 \\ V_t^{(\omega)} + \gamma^{(\omega)} 1 \end{bmatrix} = \begin{matrix} \downarrow \\ N\times 2 \\ \uparrow \end{matrix} h \begin{bmatrix} J_{te}^{(ii)} & J_{te}^{(i\omega)} \\ J_{te}^{(\omega i)} & J_{te}^{(\omega\omega)} \end{bmatrix} \quad h \begin{bmatrix} J_{te}^{(ii)} & J_{te}^{(i\omega)} \\ J_{te}^{(\omega i)} & J_{te}^{(\omega\omega)} \end{bmatrix} \quad -h \begin{bmatrix} J_{te}^{(ii)} & J_{te}^{(i\omega)} \\ J_{te}^{(\omega i)} & J_{te}^{(\omega\omega)} \end{bmatrix} \quad \begin{matrix} \leftarrow N\times 4 \\ \end{matrix} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{matrix} \downarrow \\ N\times 2 \\ \uparrow \end{matrix} \begin{bmatrix} -V_t^{(i)} + \gamma^{(i)} 1 \\ -V_t^{(\omega)} + \gamma^{(\omega)} 1 \end{bmatrix} = \begin{matrix} \downarrow \\ N\times 2 \\ \uparrow \end{matrix} -h \begin{bmatrix} J_{te}^{(ii)} & J_{te}^{(i\omega)} \\ J_{te}^{(\omega i)} & J_{te}^{(\omega\omega)} \end{bmatrix} \quad -h \begin{bmatrix} J_{te}^{(ii)} & J_{te}^{(i\omega)} \\ J_{te}^{(\omega i)} & J_{te}^{(\omega\omega)} \end{bmatrix} \quad h \begin{bmatrix} J_{te}^{(ii)} & J_{te}^{(i\omega)} \\ J_{te}^{(\omega i)} & J_{te}^{(\omega\omega)} \end{bmatrix} \quad \begin{matrix} \leftarrow N\times 4 \\ \end{matrix} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \gamma^{(i)} \\ \gamma^{(\omega)} \end{bmatrix} \begin{bmatrix} \mu^{(i)} \lambda_n^{(i)} - I^T(\lambda_t^{(i)+} + \lambda_t^{(i)-}) \\ \mu^{(\omega)} \lambda_n^{(\omega)} - I^T(\lambda_t^{(\omega)+} + \lambda_t^{(\omega)-}) \end{bmatrix} \quad \begin{matrix} \downarrow \\ N\times 1 \\ \uparrow \end{matrix} \begin{bmatrix} \mu^{(i)} & \\ & \mu^{(\omega)} \end{bmatrix} \quad \begin{bmatrix} -I^T & \\ & -I^T \end{bmatrix} \quad \begin{bmatrix} -I^T & \\ & -I^T \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\cdot \begin{bmatrix} \lambda_t^{(i)+} \\ \lambda_t^{(\omega)+} \end{bmatrix} + \begin{bmatrix} \frac{1}{h} \frac{d\phi^{(i)}(r^{(i)}(q_k))}{\| d\phi^{(i)}/dr^{(i)}(q_k) \|} + J_n^{(i)} \dot{q}_k - h J_n^{(i)} M(q_k)^{-1} C(q_k, \dot{q}_k) \\ \frac{1}{h} \frac{d\phi^{(\omega)}(r^{(\omega)}(q_k))}{\| d\phi^{(\omega)}/dr^{(\omega)}(q_k) \|} + J_n^{(\omega)} \dot{q}_k - h J_n^{(\omega)} M(q_k) C(q_k, \dot{q}_k) \end{bmatrix}$$

$$\begin{bmatrix} \lambda_t^{(i)-} \\ \lambda_t^{(\omega)-} \end{bmatrix} - \begin{bmatrix} J_t^{(i)} \dot{q}_k - h J_t^{(i)} M(q_k)^{-1} C(q_k, \dot{q}_k) \\ J_t^{(2)} \dot{q}_k - h J_t^{(2)} M(q_k)^{-1} C(q_k, \dot{q}_k) \\ J_t^{(i)} \dot{q}_k - h J_t^{(i)} M(q_k)^{-1} C(q_k, \dot{q}_k) \\ J_t^{(2)} \dot{q}_k - h J_t^{(2)} M(q_k)^{-1} C(q_k, \dot{q}_k) \end{bmatrix} \ge 0$$

$$\begin{bmatrix} \gamma^{(i)} \\ \gamma^{(\omega)} \end{bmatrix} \quad \begin{bmatrix} 0 \end{bmatrix}$$